WORKED SOLUTION

STEP II 2011, Question 13

Probability and Statistics • Measures of Skewness

STEP II 2011, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s2-q13

The key idea

Skewness is about lopsidedness, and the three measures here detect it in three different ways: through the third central moment, through where the median sits between two deciles, and through the gap between mean and median. For the density $2x$ on $[0,1]$ everything is explicit: the moments are one-line integrals, and the cumulative distribution function is x^2, so its inverse is just a square root. The only real work is careful arithmetic with surds, and a clean way to compare them at the end.

Solution

What skewness measures. Skewness measures the *asymmetry* of a distribution about its centre. A symmetric distribution has zero skewness. If the longer tail is on the right, the skewness is positive; if the longer tail is on the left, it is negative.

Part (i)

Expand the cube and use linearity of expectation, remembering that μ is a constant:

$$\begin{aligned}\mathrm{E}((X-\mu)^3) &= \mathrm{E}(X^3) - 3\mu\,\mathrm{E}(X^2) + 3\mu^2\,\mathrm{E}(X) - \mu^3 \\ &= \mathrm{E}(X^3) - 3\mu(\sigma^2+\mu^2) + 3\mu^3 - \mu^3 \\ &= \mathrm{E}(X^3) - 3\mu\sigma^2 - \mu^3,\end{aligned}$$

using $\mathrm{E}(X) = \mu$ and $\mathrm{E}(X^2) = \sigma^2 + \mu^2$ (the usual formula for the variance, rearranged). Dividing by σ^3 gives the required expression for γ.

The distribution with density $2x$. For $k \geq 0$,

$$\mathrm{E}(X^k) = \int_0^1 x^k \cdot 2x \,\mathrm{d}x = \frac{2}{k+2}.$$

So $\mu = \frac{2}{3}$, $\mathrm{E}(X^2) = \frac{1}{2}$ and $\mathrm{E}(X^3) = \frac{2}{5}$. Hence

$$\sigma^2 = \tfrac{1}{2} - \tfrac{4}{9} = \tfrac{1}{18}, \qquad \sigma = \frac{1}{3\sqrt{2}}, \qquad \sigma^3 = \frac{1}{27 \cdot 2\sqrt{2}} = \frac{1}{54\sqrt{2}}.$$

The numerator is

$$\mathrm{E}(X^3) - 3\mu\sigma^2 - \mu^3 = \frac{2}{5} - 3 \cdot \frac{2}{3} \cdot \frac{1}{18} - \frac{8}{27} = \frac{2}{5} - \frac{1}{9} - \frac{8}{27} = \frac{54 - 15 - 40}{135} = -\frac{1}{135}.$$

Therefore

$$\gamma = -\frac{1}{135} \cdot 54\sqrt{2} = -\frac{54\sqrt{2}}{135} = -\frac{2\sqrt{2}}{5},$$

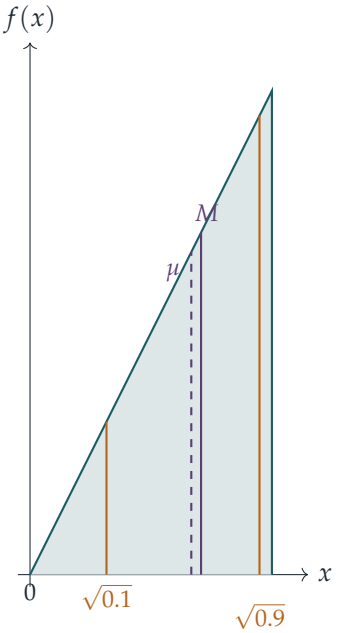
as required. The negative sign makes sense: the density piles up near $x = 1$, leaving a long tail towards 0 on the left.

Part (ii)

For $0 \le x \le 1$ the cumulative distribution function is

$$\mathrm{F}(x) = \int_0^x 2t \, \mathrm{d}t = x^2,$$

which is increasing on $[0, 1]$, so for $0 \le p \le 1$ its inverse is $\mathrm{F}^{-1}(p) = \sqrt{p}$.



Decile skewness. Multiplying the numerator and denominator of D by $\sqrt{10}$ clears all the surds except one:

$$D = \frac{\sqrt{\frac{9}{10}} - 2\sqrt{\frac{1}{2}} + \sqrt{\frac{1}{10}}}{\sqrt{\frac{9}{10}} - \sqrt{\frac{1}{10}}} = \frac{3 - 2\sqrt{5} + 1}{3 - 1} = \frac{4 - 2\sqrt{5}}{2} = 2 - \sqrt{5},$$

since $\sqrt{10}\cdot\sqrt{\frac{1}{2}}=\sqrt{5}$. This is what we had to show.

Pearson skewness. The median satisfies $\mathrm{F}(M)=\frac{1}{2}$, so $M=\frac{1}{\sqrt{2}}$. Using $\frac{1}{\sigma}=3\sqrt{2}$,

$$P=3\Big(\frac{2}{3}-\frac{1}{\sqrt{2}}\Big)\cdot 3\sqrt{2}=9\sqrt{2}\cdot\frac{2}{3}-9=6\sqrt{2}-9.$$

Numerically, $D\approx -0.236$, $P\approx -0.515$ and $\gamma\approx -0.566$, so the ordering looks right; now we prove it.

Ask yourself How do we compare numbers like $2-\sqrt{5}$ and $6\sqrt{2}-9$ without a calculator?

Rearrange so that each side is positive, then square: for positive numbers, $a<b$ if and only if $a^2<b^2$. Each step below is reversible, so a true final statement proves the first one.

$D>P$**.** We need $2-\sqrt{5}>6\sqrt{2}-9$, that is, $\sqrt{5}+6\sqrt{2}<11$. Both sides are positive, so this is equivalent to

$$\begin{aligned}(\sqrt{5}+6\sqrt{2})^2=77+12\sqrt{10}<121 &\iff 12\sqrt{10}<44\\ &\iff 3\sqrt{10}<11\\ &\iff 90<121,\end{aligned}$$

where the last step squares two positive numbers again. Since $90<121$ is true, $D>P$.

$P>\gamma$**.** We need $6\sqrt{2}-9>-\frac{2\sqrt{2}}{5}$. Multiplying by 5 and rearranging, this is $32\sqrt{2}>45$. Both sides are positive, so it is equivalent to $32^2\cdot 2>45^2$, that is, $2048>2025$, which is true. So $P>\gamma$.

Hence $D>P>\gamma$. All three measures are negative and agree that the distribution is skewed to the left, but they disagree about how much.

Checks

- The identity in (i) gives 0 for any symmetric distribution: for example, if X is ± 1 with equal probability then $\mu=0$, $\sigma=1$ and $\mathrm{E}(X^3)=0$.
- Numerically, a midpoint-rule evaluation of $\mathrm{E}\big((X-\mu)^3\big)/\sigma^3$ gives -0.56569, matching $-2\sqrt{2}/5$.
- Simulating $X=\sqrt{U}$ with U uniform on $[0,1]$ (which has $\mathrm{F}(x)=x^2$) gives a sample median close to 0.707 and deciles close to 0.316 and 0.949, as in the sketch.
- The mean $\frac{2}{3}$ lies to the left of the median $\frac{1}{\sqrt{2}}\approx 0.707$, which is typical of a left-skewed distribution and is why $P<0$.

Common slips

- Writing $\sigma^3 = (\frac{1}{18})^3$ instead of $(\frac{1}{18})^{3/2}$: it is the standard deviation, not the variance, that is cubed.
- Squaring an inequality when one side may be negative. Rearrange first so that both sides are positive.
- Using $\mathrm{E}(X^2) = \frac{1}{2}$ where $\sigma^2 = \frac{1}{18}$ is needed, in either the formula for γ or the denominator of P.

Beyond STEP

The quantity $\mathrm{E}\big((X-\mu)^3\big)$ is the third *central moment*, and the expansion in (i) is the start of the relationship between moments and *cumulants*, which add up nicely for sums of independent random variables. Quantile-based measures such as D are prized in statistics because they are *robust*: a single wild observation barely moves a decile but can change a third moment dramatically. Grimmett and Welsh, *Probability: An Introduction*, develops moments and moment generating functions at the right level.

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